Sectional, assembly & exploded views

GRAPHICS WITH MR KELLY

National 5 · Drawing & spatial awareness · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ read and draw a sectional view with a labelled cutting plane
- ☐ apply hatching correctly, leaving webs, shafts, nuts and bolts unhatched
- ☐ describe good and poor practice in an exploded view

I DO — Worked example

Read how a full-mark answer is built

An exploded pictorial of a desk lamp is shown.

1 Describe two features of good practice in an exploded view. (2 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Parts are spaced so none overlap
- ✓ All parts can be seen

WE DO — Guided practice

Use the starters and word bank to help you

A sectional assembly of a pulley shows a casing, a shaft, a nut and bolt, and a strengthening web.

2 Explain which parts should be hatched and which should not. (3 marks)

Sentence starters

One reason is … because …
Another reason is … so …
Also, … which means …

Word bank

hatched · casing · material · spacings · adjacent · lengthways · directions · different

Tip: 3 marks = 3 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Hatching lines are normally drawn at...
☐ 30° ☐ 45° ☐ 60° ☐ 90°

b) Which of these is NOT hatched when cut lengthways?
☐ A casing ☐ A bolt ☐ A base plate ☐ A housing

c) Which is a feature of a good exploded view?
☐ Parts overlap to save space ☐ Parts are aligned along assembly lines ☐ Parts are shown in random order ☐ Hidden detail on every part

4 Key terms: write the letter of the matching description next to each term.

1. **Hatching**	___	**A** Webs, ribs, shafts, nuts, bolts, screws, washers, pins and threads cut lengthways.
2. **Which parts are not hatched?**	___	**B** Thin 45° lines on cut surfaces of a sectional view.
3. **SECTION A–A**	___	**C** The label for the view taken at cutting plane A–A.

5 Explain what is meant by ‘Good exploded view’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?